C

MULTIVARIABLE CALCULUS • UNIT 5.01

Triple Integrals, Cylindrical & Spherical Coordinates

PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Triple Integrals, Cylindrical & Spherical Coordinates Practice Exercises

1 MCQ

Evaluate $\iiint_B xyz\,dV$ over the box $B = [0,3] \times [0,1] \times [0,3]$.

A $\frac{81}{4}$

B $\frac{81}{16}$

C $\frac{81}{8}$

D $\frac{153}{8}$

2 MCQ

Evaluate $\iiint_B x + y + z\,dV$ over the box $B = [0,3] \times [0,2] \times [0,1]$.

A 36

B 9

C 18

D 24

3 MCQ

Evaluate $\iiint_B xyz\,dV$ over the box $B = [0,2] \times [0,1] \times [0,2]$.

A 4

B 1

C 6

D 2

4 MCQ

Evaluate $\iiint_B yz^2\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

A 9

B $\frac{9}{4}$

C $\frac{15}{2}$

D $\frac{9}{2}$

5 MCQ

Evaluate $\iiint_B x^2z\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

A 3

B $\frac{3}{4}$

C $\frac{3}{2}$

D $\frac{9}{2}$

6 MCQ

Evaluate $\iiint_B xy + z\,dV$ over the box $B = [0,1] \times [0,1] \times [0,3]$.

A $\frac{21}{4}$

B $\frac{21}{2}$

C $\frac{21}{8}$

D $\frac{33}{4}$

7 MCQ

Evaluate $\iiint_B xy + z\,dV$ over the box $B = [0,1] \times [0,3] \times [0,3]$.

A $\frac{81}{2}$

B $\frac{81}{8}$

C $\frac{81}{4}$

D $\frac{117}{4}$

8 MCQ

Evaluate $\iiint_B x + y + z\,dV$ over the box $B = [0,3] \times [0,3] \times [0,1]$.

A 63

B $\frac{63}{2}$

C $\frac{63}{4}$

D $\frac{81}{2}$

9 MCQ

Consider the vector field $\mathbf{u}(s, y, z) = \langle 22s,\ 7y,\ 7z \rangle$ on R^3. Compute the scalar divergence $\nabla \cdot \mathbf{u}$.

A 40

B 29

C 58

D 36

10 MCQ

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_{\alpha}^{\beta} \int_{\gamma}^{\delta} \int_{\varrho_1}^{\varrho_2} \varrho^2 \sin\phi \, d\varrho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

A $\frac{\pi}{2}$

B π

C 0

D 2π

11 MCQ

A solid region E is defined by the inequalities $x^2 + y^2 \leq 16$, $0 \leq z \leq 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$, and $0 \leq z \leq H$. What is the value of R?

A 2

B 4

C 8

D 16

12 MCQ

Consider the solid region E in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4-r^2} r\,dz\,dr\,d\theta$. Find the value of the parameter a.

A 4

B 2

C 16

D 8

13 MCQ

Let the multivariable function u be defined by $u(s, y) = 14s^2y + 24y^2$. Compute the partial derivative $\frac{\partial u}{\partial s}(6, 10)$.

A 1680

B 1656

C 1684

D 1694

14 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \leq 4$.

A 4π

B $\frac{32\pi}{3}$

C $\frac{16\pi}{3}$

D $\frac{8\pi}{3}$

15 MCQ

Use spherical coordinates to find the volume of the first-octant part of the ball $x^2 + y^2 + z^2 \leq 16$.

A $\frac{32\pi}{3}$

B $\frac{16\pi}{3}$

C $\frac{256\pi}{3}$

D $\frac{64\pi}{3}$

16 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{2}{4}\sqrt{x^2 + y^2} \leq z \leq 2$.

A $\frac{8\pi}{3}$

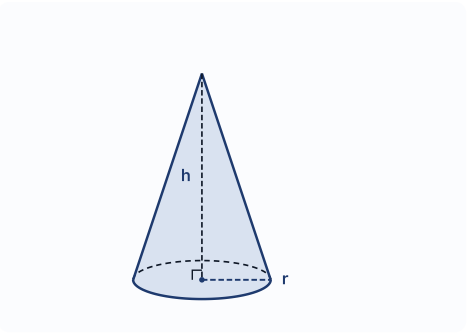
B 32π

C $\frac{32\pi}{3}$

D 16π

17 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{1}{3}\sqrt{x^2+y^2} \le z \le 1$.



A 9π

B 3π

C $\frac{9\pi}{2}$

D π

18 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{4}\sqrt{x^2+y^2} \le z \le 4$.

A $\frac{16\pi}{3}$

B 64π

C $\frac{64\pi}{3}$

D 32π

19 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2+y^2+z^2 \le 9$.

A $\frac{27\pi}{2}$

B 18π

C 36π

D 9π

20 MCQ

Use spherical coordinates to find the volume of the upper half of the ball $x^2+y^2+z^2 \le 16$.

A $\frac{256\pi}{3}$

B $\frac{64\pi}{3}$

C 32π

D $\frac{128\pi}{3}$

21 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{3}{4}\sqrt{x^2+y^2} \le z \le 3$.

A 16π

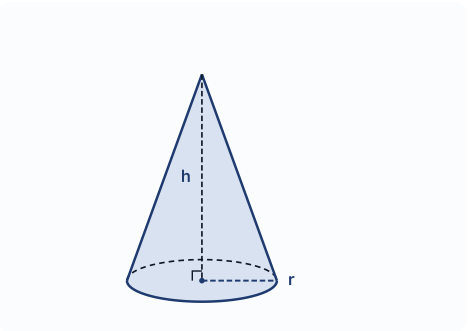
B 48π

C 24π

D 4π

22 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{1}\sqrt{x^2+y^2} \le z \le 4$.



A 4π

B 2π

C $\frac{4\pi}{3}$

D $\frac{8\pi}{3}$

23 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{2}{3}\sqrt{x^2+y^2} \le z \le 2$.

A 18π

B 9π

C 6π

D 2π

24 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{4}{2}\sqrt{x^2+y^2} \le z \le 4$.

A $\frac{16\pi}{3}$

B $\frac{8\pi}{3}$

C 16π

D 8π

25 MCQ

Use cylindrical coordinates to find the volume of the cone $\frac{3}{2}\sqrt{x^2+y^2} \le z \le 3$.

A 4π

B 12π

C 6π

D 2π